

6 Dietary antioxidants may reduce the risk of cardiovascular disease (CVD).

(a) Explain how dietary antioxidants reduce the risk of CVD.

(3)

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- (b) Chocolate contains high concentrations of a group of antioxidants called flavonoids. It has been suggested that eating chocolate could reduce the risk of CVD.

The table shows some information about two types of chocolate.

Type of chocolate	Mass of flavonoids / mg per 100 g of chocolate	Energy content / kJ per 100 g of chocolate
milk chocolate	70	2345
dark chocolate	170	1800

- (i) Explain whether eating dark chocolate is likely to reduce the risk of CVD more than eating milk chocolate. Use the information in the table to support your answer.

(4)

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- (ii) The table gives some information about ingredients in three different brands of milk chocolate and dark chocolate.

Ingredient	Percentage of ingredient in milk chocolate (%)			Percentage of ingredient in dark chocolate (%)		
	brand 1	brand 2	brand 3	brand 1	brand 2	brand 3
cocoa mass	16	16	16	40	40	40
sugar	40	40	40	50	50	50
milk fat	0	3	0	0	3	1
whole milk powder	20	20	20	0	0	0

Use the information in both tables to identify which ingredient contains the most flavonoids.

(1)

- ☐ **A** cocoa mass
- ☐ **B** milk fat
- ☐ **C** sugar
- ☐ **D** whole milk powder

- (iii) Describe how a study could be designed to collect valid and reliable data on the effects of eating chocolate on the risk of CVD.

(3)

(Total for Question 6 = 11 marks)



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